

# STAT425

Ivy Gu & Jayden Zhou

2025-04-20

Read in data

```
library(MASS)
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 4.3.3
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.3.3
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.Date, as.Date.numeric
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.3.3
```

```
climate = read.csv("DailyDelhiClimateTrain.csv")
```

remove outliers based on IQR

```
remove_outliers_iqr <- function(df, columns) {
  for (col in columns) {
    Q1 <- quantile(df[[col]], 0.25, na.rm = TRUE)
    Q3 <- quantile(df[[col]], 0.75, na.rm = TRUE)
    IQR_value <- Q3 - Q1
    lower_bound <- Q1 - 1.5 * IQR_value
    upper_bound <- Q3 + 1.5 * IQR_value
    df <- df[df[[col]] >= lower_bound & df[[col]] <= upper_bound, ]
  }
}
```

```

    }
    return(df)
  }

climate_clean <- remove_outliers_iqr(climate, c("humidity", "wind_speed", "meanpressure", "meantemp"))

```

Groupby -> Year\_Month

```

library(tidyr)

# divided into three columns: year, month, day
climate_clean <- separate(climate_clean, col = date, into = c("year", "month", "day"), sep = "-")

# then create year_month
climate_clean$year_month <- paste(climate_clean$year, climate_clean$month, sep = "-")

library(ggplot2)
library(dplyr)

```

```

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:car':
##
##      recode

## The following object is masked from 'package:MASS':
##
##      select

## The following objects are masked from 'package:stats':
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union

```

```

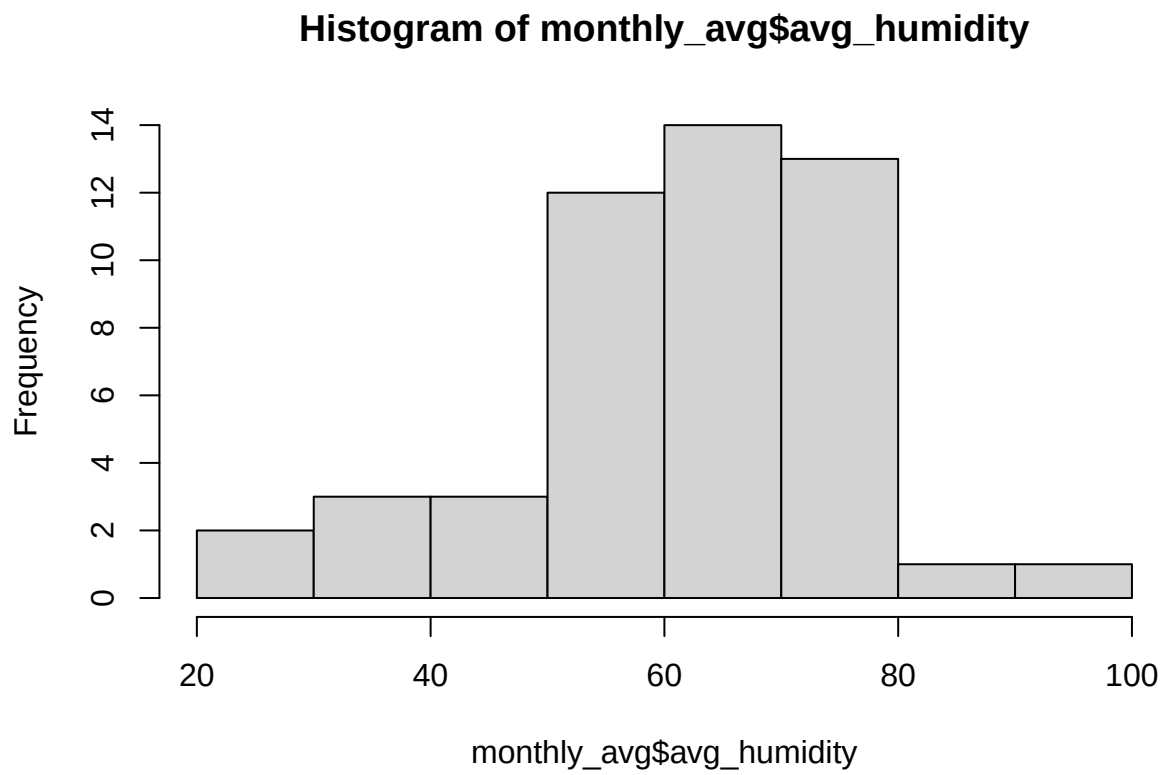
library(dplyr)

monthly_avg <- climate_clean %>%
  group_by(year_month) %>%
  summarise(
    avg_temp = mean(meantemp, na.rm = TRUE),
    avg_humidity = mean(humidity, na.rm = TRUE),
    avg_pressure = mean(meanpressure, na.rm = TRUE),
    avg_wind_speed = mean(wind_speed, na.rm = TRUE)
  )

```

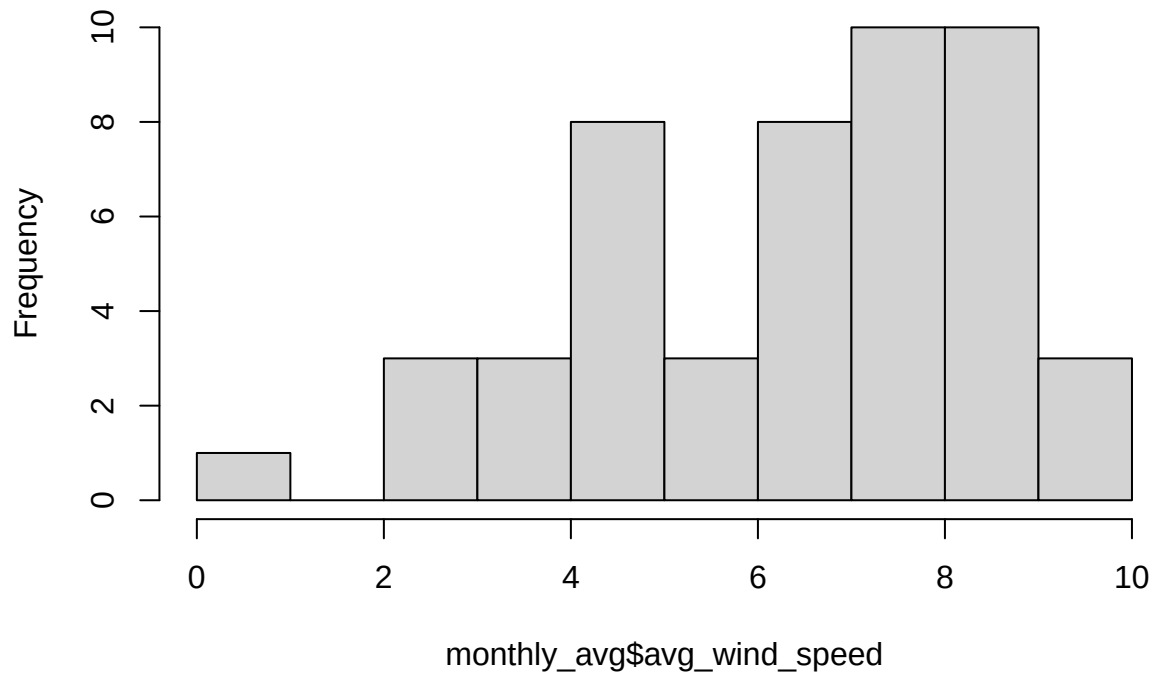
Histogram

```
hist(monthly_avg$avg_humidity)
```



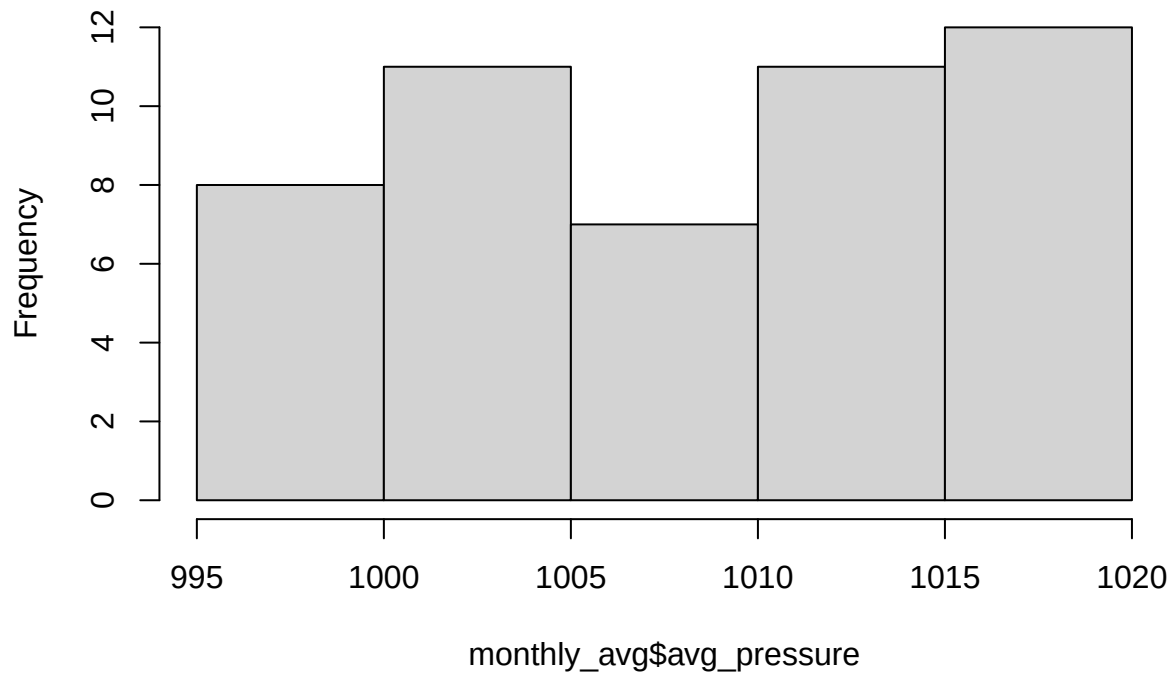
```
hist(monthly_avg$avg_wind_speed)
```

**Histogram of monthly\_avg\$avg\_wind\_speed**



```
hist(monthly_avg$avg_pressure)
```

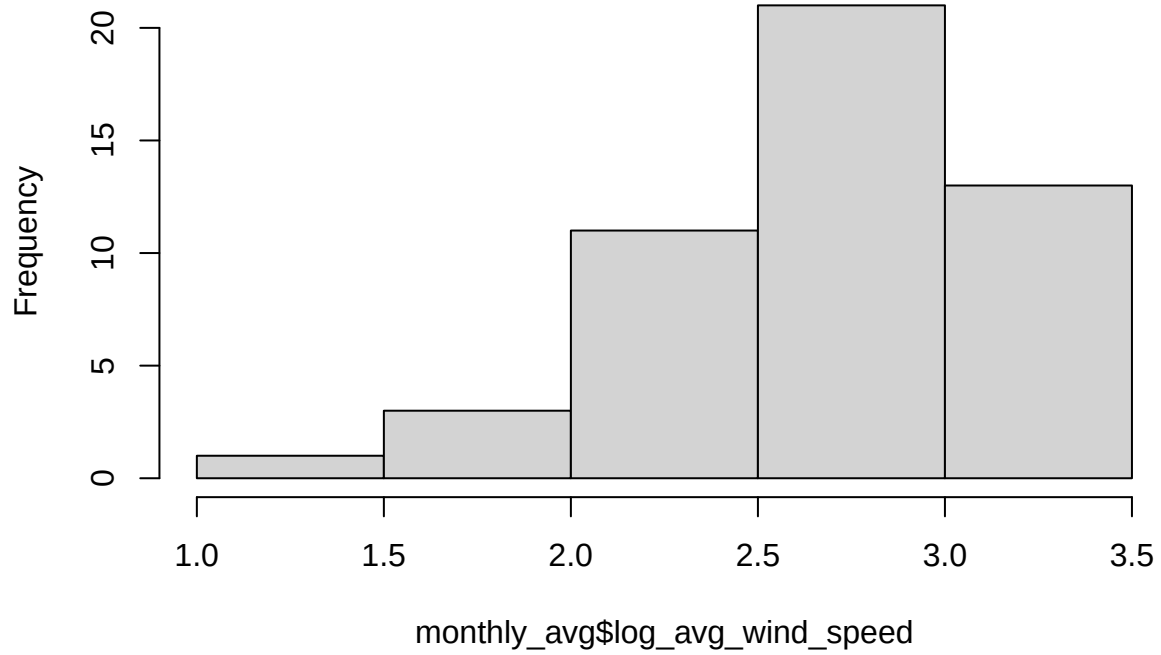
**Histogram of monthly\_avg\$avg\_pressure**



```
monthly_avg$log_avg_wind_speed = sqrt(monthly_avg$avg_wind_speed+1)
monthly_avg$log_avg_pressure = log(monthly_avg$avg_pressure+1)

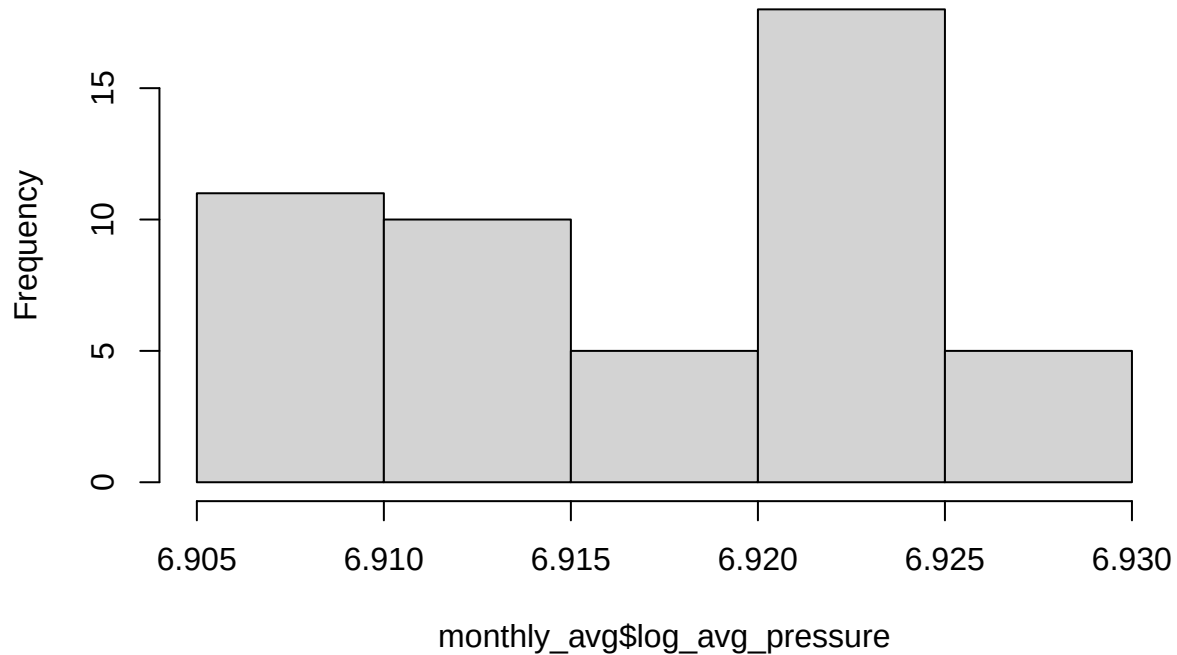
hist(monthly_avg$log_avg_wind_speed)
```

**Histogram of monthly\_avg\$log\_avg\_wind\_speed**



```
hist(monthly_avg$log_avg_pressure)
```

## Histogram of monthly\_avg\$log\_avg\_pressure



```
climate_mod = lm(avg_temp~avg_humidity+avg_pressure+avg_wind_speed, data = monthly_avg)
monthly_avg$residual <- residuals(climate_mod)
monthly_avg$year_month_date <- as.Date(paste0(monthly_avg$year_month, "-01"))
```

Collinearity Check

```
vif(climate_mod)
```

```
## avg_humidity avg_pressure avg_wind_speed
## 1.767689 1.682539 2.446189
```

```
x = model.matrix(climate_mod)[,-1]
x = x - matrix(apply(x, 2, mean), 49, 3, byrow = TRUE)
x = x / matrix(apply(x, 2, sd), 49, 3, byrow = TRUE)
apply(x, 2, mean)
```

```
## avg_humidity avg_pressure avg_wind_speed
## -2.537653e-16 4.622153e-15 -1.450087e-16
```

```
apply(x, 2, var)
```

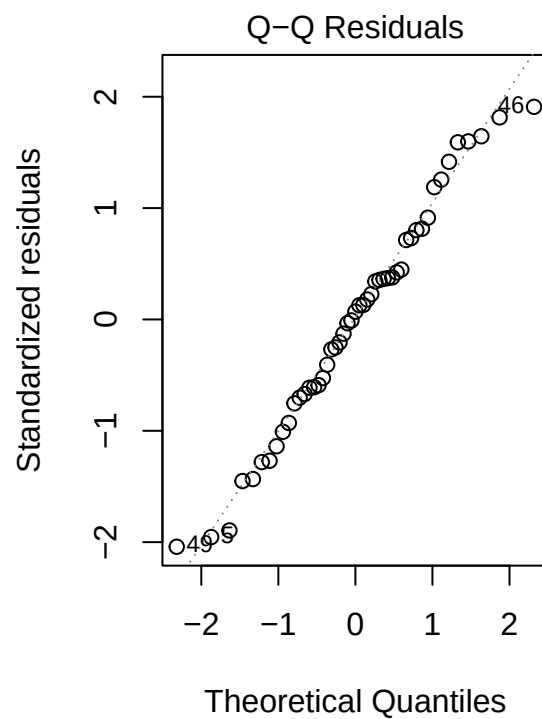
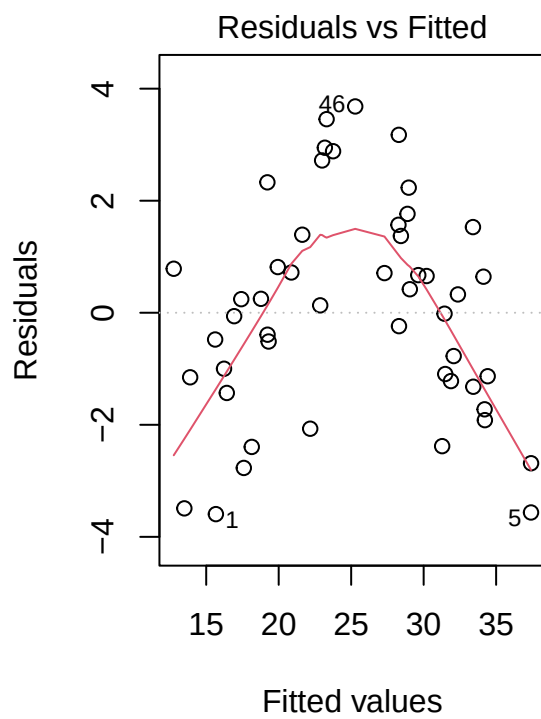
```
## avg_humidity avg_pressure avg_wind_speed
## 1 1 1
```

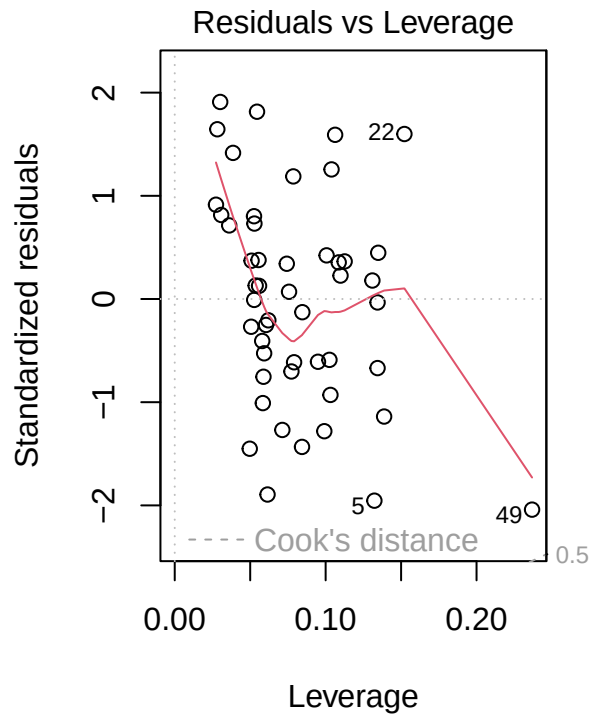
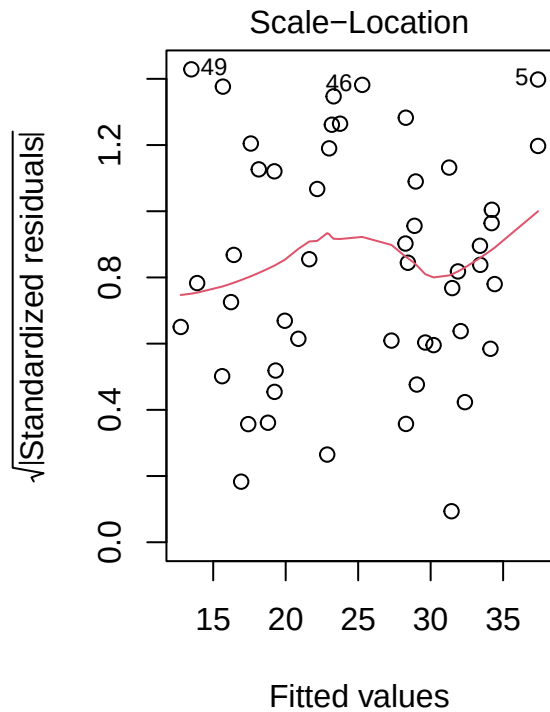
```
e = eigen(t(x) %*% x)
sqrt(e$val[1]/e$val)
```

```
## [1] 1.000000 1.927291 2.822880
```

Diagnostics

```
par(mfrow = c(1,2))
plot(climate_mod)
```





```
par(mfrow = c(1,1))
```

```
bptest(climate_mod)
```

```
##
## studentized Breusch-Pagan test
##
## data: climate_mod
## BP = 4.4619, df = 3, p-value = 0.2157
```

```
shapiro.test(residuals(climate_mod))
```

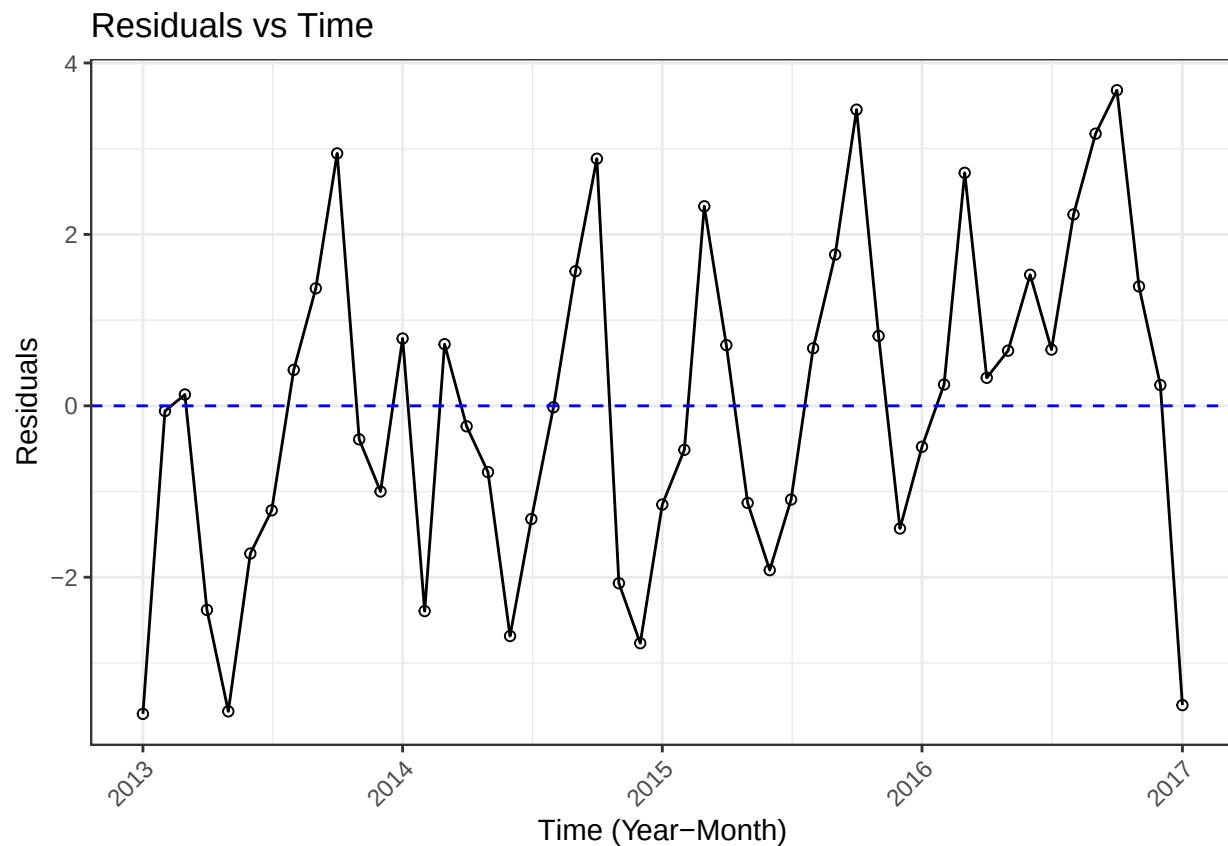
```
##
## Shapiro-Wilk normality test
##
## data: residuals(climate_mod)
## W = 0.98032, p-value = 0.5787
```

Residuals vs Time”

```
library(ggplot2)
```

```
ggplot(monthly_avg, aes(x = year_month_date, y = residual)) +
  geom_point(shape = 1) +
```

```
geom_line() +
geom_hline(yintercept = 0,
           color = "blue", linetype = "dashed") +
labs(title = "Residuals vs Time",
     x = "Time (Year-Month)",
     y = "Residuals") +
theme_bw() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# View(monthly_avg)
```

dwtest:auto-correlation

```
dwtest(climate_mod)
```

```
##
## Durbin-Watson test
##
## data: climate_mod
## DW = 1.0361, p-value = 6.781e-05
## alternative hypothesis: true autocorrelation is greater than 0
```

Using AR1 model

```
monthly_avg$year_month <- gsub("-", "", monthly_avg$year_month)
library(nlme)
```

```
## Warning: package 'nlme' was built under R version 4.3.3
```

```
##
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':
##
## collapse
```

```
gls_mod = gls(avg_temp ~ avg_humidity + avg_pressure + avg_wind_speed, correlation = corAR1(form = ~ as
intervals(gls_mod)
```

```
## Approximate 95% confidence intervals
##
## Coefficients:
##               lower      est.      upper
## (Intercept)  717.5724028 853.69686861 989.8213344
## avg_humidity   -0.2110633  -0.16180408 -0.1125449
## avg_pressure   -0.9453373  -0.81150490 -0.6776724
## avg_wind_speed -0.4970046  -0.08767426  0.3216560
##
## Correlation structure:
##               lower      est.      upper
## Phi1 0.2234126 0.5575585 0.7744378
##
## Residual standard error:
##               lower      est.      upper
## 1.529914 2.063042 2.781951
```

Seasons

```
monthly_avg$month <- as.numeric(monthly_avg$year_month) %% 100

monthly_avg$season <- ifelse(monthly_avg$month %in% c(12, 1, 2), "Winter",
  ifelse(monthly_avg$month %in% c(3, 4, 5), "Spring",
    ifelse(monthly_avg$month %in% c(6, 7, 8), "Summer",
      ifelse(monthly_avg$month %in% c(9, 10, 11), "Autumn", NA))))
```

ANCOVA

```
Amodel <- lm(avg_temp ~ avg_humidity + avg_pressure + avg_wind_speed + season + avg_humidity:season + a
anova(Amodel)
```

```
## Analysis of Variance Table
##
## Response: avg_temp
##               Df   Sum Sq Mean Sq  F value    Pr(>F)
```

```
## avg_humidity          1 1081.55 1081.55 561.0577 < 2.2e-16 ***
## avg_pressure          1 1296.40 1296.40 672.5132 < 2.2e-16 ***
## avg_wind_speed        1    3.45    3.45    1.7876    0.19036
## season                3    83.81    27.94    14.4919 3.425e-06 ***
## avg_humidity:season    3     4.85     1.62     0.8385    0.48249
## avg_pressure:season    3    19.06     6.35     3.2959    0.03244 *
## avg_wind_speed:season  3     1.22     0.41     0.2114    0.88775
## Residuals             33    63.61     1.93
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# deleted avg_wind_speed:season
```

```
Bmodel <- lm(avg_temp ~ avg_humidity + avg_pressure + avg_wind_speed + season + avg_pressure:season + avg_wind_speed:season, data = data)
anova(Bmodel)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_temp
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## avg_humidity    1 1081.55 1081.55 600.5194 < 2.2e-16 ***
## avg_pressure    1 1296.40 1296.40 719.8141 < 2.2e-16 ***
## avg_wind_speed   1    3.45    3.45    1.9134    0.17511
## season          3    83.81    27.94   15.5112 1.222e-06 ***
## avg_pressure:season 3    21.56     7.19    3.9899    0.01495 *
## avg_humidity:season 3     2.35     0.78    0.4353    0.72906
## Residuals       36    64.84     1.80
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# deleted avg_humidity:season
```

```
Cmodel <- lm(avg_temp ~ avg_humidity + avg_pressure + avg_wind_speed + season + avg_pressure:season, data = data)
anova(Cmodel)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_temp
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## avg_humidity    1 1081.55 1081.55 627.7902 < 2.2e-16 ***
## avg_pressure    1 1296.40 1296.40 752.5023 < 2.2e-16 ***
## avg_wind_speed   1    3.45    3.45    2.0003    0.16521
## season          3    83.81    27.94   16.2156 5.352e-07 ***
## avg_pressure:season 3    21.56     7.19    4.1711    0.01179 *
## Residuals       39    67.19     1.72
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Dmodel <- lm(avg_temp ~ avg_humidity + avg_pressure + season + avg_pressure:season + avg_wind_speed, data = data)
anova(Dmodel)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_temp
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## avg_humidity    1 1081.55 1081.55 627.7902 < 2.2e-16 ***
## avg_pressure    1 1296.40 1296.40 752.5023 < 2.2e-16 ***
## season          3   81.54   27.18  15.7758 7.153e-07 ***
## avg_wind_speed    1    5.72    5.72   3.3197 0.07613 .
## avg_pressure:season 3   21.56    7.19   4.1711 0.01179 *
## Residuals       39   67.19    1.72
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(Dmodel)
```

```
##
## Call:
## lm(formula = avg_temp ~ avg_humidity + avg_pressure + season +
##     avg_pressure:season + avg_wind_speed, data = monthly_avg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7066 -0.7733  0.1624  0.8032  2.2294
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    891.08268   114.56363    7.778 1.88e-09 ***
## avg_humidity    -0.13352    0.02988   -4.468 6.61e-05 ***
## avg_pressure    -0.84965    0.11222   -7.571 3.57e-09 ***
## seasonSpring   -246.75013   162.48392   -1.519 0.13693
## seasonSummer   -950.30640   271.72890   -3.497 0.00119 **
## seasonWinter   -240.77987   300.20503   -0.802 0.42738
## avg_wind_speed    0.23383    0.19350    1.208 0.23416
## avg_pressure:seasonSpring  0.24239    0.16114    1.504 0.14058
## avg_pressure:seasonSummer  0.94749    0.27162    3.488 0.00122 **
## avg_pressure:seasonWinter  0.23364    0.29560    0.790 0.43407
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.313 on 39 degrees of freedom
## Multiple R-squared:  0.9737, Adjusted R-squared:  0.9676
## F-statistic: 160.4 on 9 and 39 DF, p-value: < 2.2e-16
```

```
ANCOVA_model <- lm(avg_temp ~ avg_humidity + avg_pressure + season + avg_pressure:season, data = monthl
anova(ANCOVA_model)
```

```
## Analysis of Variance Table
```

```
##
## Response: avg_temp
##              Df Sum Sq Mean Sq F value    Pr(>F)
## avg_humidity    1 1081.55 1081.55 620.6486 < 2.2e-16 ***
## avg_pressure    1 1296.40 1296.40 743.9421 < 2.2e-16 ***
## season          3   81.54   27.18  15.5963 7.2e-07 ***
## avg_pressure:season 3   24.76    8.25   4.7364 0.006407 **
## Residuals       40   69.70    1.74
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rmse <- function(x,y) sqrt(mean((x - y)^2))  
rmse(fitted(gls_mod), monthly_avg$avg_temp)
```

```
## [1] 1.916042
```

```
rmse(fitted(ANCOVA_model), monthly_avg$avg_temp)
```

```
## [1] 1.192702
```

```
rmse(fitted(climate_mod), monthly_avg$avg_temp)
```

```
## [1] 1.876572
```